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RESEARCH ARTICLE

The ethics of care and the tragedy of the commons

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Abstract
In this paper, we argue that the ethics of care can offer solutions to diverse tragedies of the commons. Building on Hayek's (Law, legislation and liberty, Abingdon, Routledge, 1982) theory on the emergence of rules and norms, we outline the spheres of application of the ethics of care and the ethics of justice and outline how the two interact. We apply our analysis of the interaction of the ethics of care and the ethics of justice to the problem of the tragedy of the commons. The morality of care allows people to establish rules in small groups and to ensure the compliance with theoretically defined rules in the Great Society, thus potentially avoiding any tragedy of the commons in either realm. We illustrate our argument with two examples of the tragedy of the commons: fractional-reserve banking and environmental tragedies.

Keywords Ethics of care · Tragedy of the commons · Limitations of knowledge · Fractional-reserve banking · Environmental management

JEL Classification E42 · K11 · P48

1 Introduction

The ethics of care is an approach to moral issues based on the feminist idea of care. In one of the most comprehensive works written on the ethics of care, Held (2006, 10) emphasizes that the ethics of care is focused on the "moral salience of attending to and meeting the needs of the particular others for whom we take responsibility." Proponents of an ethics of care regard emotions such as empathy, sympathy, and sensitivity as essential for moral guidance and reject abstract reasoning as the only way to deal with a moral problem. The approach is meant to improve relationships

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within certain domains, such as the family or networks of friends, in which emotions are the primary moral guide.

Since the emergence of the ethics of care in the 1980s, it has been continually compared with the ethics of justice. Gilligan (1982), who first used the term "the ethics of care," expounded the idea of an ethic of care in contrast to the idea of an ethic of justice. So did other renowned authors such as Noddings (1984), Tronto (1993), Ruddick (1995), and Held (2006). In fact, the ethics of care appeared to be an alternative moral approach to dominant branches of ethics such as utilitarianism, Kantian ethics, and virtue ethics. According to care ethicists, the ethics of justice, which is concerned with the rational, self-interested, and autonomous individual who has to observe universal, impersonal, impartial, and abstract moral principles, cannot provide complete guidance for human behavior. Are the members of a family guided continually by such principles? Are not many decisions personal, partial, and based on emotions? The ethics of care responds to these questions by supporting care as a practice and as a value (Held 2006)—that is, not only to describe specific actions that are not guided by universal and abstract rules but also to advocate care as a moral goal that acknowledges responsiveness, sensitivity, empathy, cooperation, trust, actions directed by emotions, and relatedness.

The contrast between the ethics of care and the ethics of justice is summed up by Held (2006):

> An ethic of justice focuses on questions of fairness, equality, individual rights, abstract principles, and the consistent application of them. An ethic of care focuses on attentiveness, trust, responsiveness to need, narrative nuance, and cultivating caring relations. . . . Whereas justice protects equality and freedom, care fosters social bonds and cooperation (15).
>
> Care valued relationships between persons and empathetic understanding; justice valued rational action in accord with abstract principles (62).

However, the two ethical approaches are not incompatible. Many ethicists have written on the relationship between them. Some prioritize care over justice (Held 2006)–some even to the point of thinking care should replace justice as the central concept of morality (Noddings 1984)–and others vice versa (Barry 1995). A common thread is that care and justice should complement each other. The question, then, is to delimit concrete domains for the application of each ethic and to describe the interaction of these ethics. Noddings (1984) and others, in early work on the subject, posit that the ethics of care should be confined to the private sphere, whereas the ethics of justice should prevail in public realm. Tronto (1993) proposes to expand the realm of the ethics of care to the political domain and thus end the private/public dichotomy that characterizes some versions of the ethics of justice, a proposal that was later seconded by other authors such as Bubeck (1995) and Kittay (1999). Held (2006, 69), however, states: "We need different moral approaches for different domains." She thinks that justice should be a principal value in the field of law, whereas care should be the primary value in the sphere of the family. However, an ethic of care may be suitable in public contexts

as well. In fact, Held (2006) affirms that, provided that people are interdependent, the whole social fabric is sustained on care.

From this view of the ethics of care a vast array of applications to diverse fields of study have been developed: sociology (England 2005), business ethics (Liedtka 1996; Solomon 1998; White 1992), philosophy (Held 2006; Pulcini 2013), political science (Tronto 1993), social psychology (Paillé et al. 2016) or even neuroscience (Colombo 2014). Among these fields is economics as well. Contrary to traditional views on economics from the perspective of justice, some authors have introduced new analyses from the perspective of care (Nelson 2011; Nelson and England 2002; van Staveren 2005, 2013). In this paper, we develop an application of the ethics of care to situations in which property rights are not well defined or, despite being well defined, are not enforced in practice, which results in the tragedy of the commons. We reckon that putting into practice the values advocated by the care ethicists can solve the economic problems of the tragedy of the commons.

In order to accomplish our task, we first present a new interpretation of the interaction between the ethics of justice and the ethics of care grounded in the works of Friedrich A. Hayek and his conception of the limitations of knowledge. Secondly, we argue that an ethic of care can solve the problem of the tragedy of the commons. Finally, we provide two illustrations of the problem—fractional-reserve banking and environmental tragedies—and discuss how the ethics of care can cope with these two problems and lead to a beneficial outcome.

2 Ethics of care and ethics of justice: an issue of knowledge

As said above, several authors have tried to demarcate the domains of application of the ethics of care and the ethics of justice and to describe their interaction. This pursuit is relevant to our own insofar as our objective is to apply the ethics of care to specific realms of business. Is business in the sphere of the ethics of care? What business activities and topics lie within that sphere? If so, we contribute a distinctive explanation of how the two ethics interact and what their domains are.

In F. A Hayek's work, limitations of what people can know constitute the standard and basic notion on which economic, social, and political theory is built. Hayek (1982) remarks that the increased scale of modern society, which he calls the Great Society, rests on an immense volume of factual knowledge. This vast amount of knowledge concerns the circumstances of time and place that influence the actions of the members of society. No individual can acquire all of this knowledge, not only because of our limited cognitive ability but also because of the private and tacit nature of this knowledge. In current societies, everyone interacts directly and indirectly with strangers. This reality entails that individuals continuously live and act in irremediable ignorance about most of the concrete facts that inform the complex processes of society. Because of this ignorance, Hayek (1982, 13) asserts that.

> most of the rules of conduct which govern our actions, and most of the institutions which arise out of this regularity [including morality], are adaptations to the impossibility of anyone taking conscious account of all the particular facts

> which enter into the order of society. We shall see, in particular, that the possibility of justice rests on this necessary limitation of our factual knowledge.

According to Hayek, so long as we cannot know the concrete facts, reasons, or emotions that motivate the actions of third parties—in the case of the Great Society, we must refer to millions of third parties—we cannot rationally establish an ethics that accounts for the tremendous amount of factual knowledge dispersed across millions of minds. It is impossible to rationally determine which norms or rules members of the Great Society must obey.[1] Such norms are chosen in a long process of trial and error, where the best norms survive as long as they foster the development and expansion of society:

> Most rules of conduct are thus not derived by an intellectual process from the knowledge of the facts of the environment, but constitute the only adaptation of man to these facts which we have achieved, a knowledge of them of which we are not aware and which does not appear in our conceptual thought, but which manifests itself in the rules which we obey in our actions (Hayek 1982, 21).

Great Society rules have not been designed for any individuals or groups of individuals; hence, they are end independent and cannot emerge from partial and personal decisions.[2] That is why we need an abstract ethical framework that enables us to coordinate and make the most of our dispersed factual knowledge. Rules of conduct serve to facilitate cooperation by reducing—not eliminating—uncertainty. The most intricate ethical framework, made up of various rules of conduct, cannot determine concrete actions positively but must demarcate the limits of individual action in a negative form: rules of conduct must not tell us what to do but what not to do, precisely because of limitations of what we can know (Hayek 1982). This is the approach associated with the concept of negative liberty. Consequently, the abstract, impartial, and universal character of the ethics of justice is a consequence of the limitations of knowledge.

As Hayek (1982) points out, our knowledge is very limited in the Great Society. However, this is not so in small groups.[3] In a small group, individuals can know more about the concrete emotions, motives, reasons, and circumstances of other members of the group; they do not live in a state of irremediable ignorance. Consequently, according to Hayek, they do not need abstract and universal rules but

[1] Note that other proponents of a free market system differ from Hayek (1982). Rothbard (1982) rationally establishes an ethics of capitalism.

[2] On the evolutionary origin of law, see also Leoni (1991). An excellent presentation of the evolutionary origin of rules is also provided by van den Hauwe (1998).

[3] When we refer to the Great Society and small groups, we do not mean to refer to an exact amount of people. We use these terms as imaginary constructions to simplify the analysis. We can differentiate a small group from the Great Society based on the fact that the set of rules governing the small group can be rationally designed, whereas in the Great Society, the amount of dispersed factual knowledge makes it impossible to rationally design the governing set of rules.

can design their own rules and customs according to their common ends.[4] While the negative rights of the Great Society demarcate a sphere of action in which an individual may act freely without interference, in a small group the emphasis shifts to the positive obligations toward other members of the group. As the small group grows into the Great Society, people need to make the duties they owe to all others less onerous. The greater the group, the greater the likelihood that the common knowledge of the group will concern abstract features of actions and things. The concrete norms of the small group evolve into abstract and universal rules for civilization as a result of the increase of unknown factual information that underlies the whole of social cooperation. Another way to state the fact is that when small groups evolve into larger groups, more and more positive rights disappear and make place for the negative rights associated with the Great Society.

From this Hayekian analysis, we can affirm that caring, as the central piece of the ethics of care, is a practice and value of small groups. So long as caring emerges from face-to-face interactions in which the carer can meet the needs of the cared-for (Bubeck 1995), at least the carer possesses intricate knowledge about the situation and circumstances of the cared-for. Moreover, as the ethics of care rejects the universal and impartial character of the ethics of justice and advocates emotions such as empathy or sensitivity instead of abstract principles (Held 2006), it comes gradually closer to the features of small-group morality defined by Hayek (1982).

That being so, we can demarcate the domains of the ethics of care and the ethics of justice. The ethics of care, conceived as practices, values, and norms, is confined to the small group because caring for someone or something necessitates that one knows concrete facts which makes it impossible for the ethics of care to be applied to the Great Society as the primary moral guide, given that current society rests on a vast amount of factual knowledge that is incomprehensible by a single or few minds. The spheres of application of the ethics of care are those in which individuals can know each other and the circumstances of each other's lives. In contrast, the ethics of justice is indispensable in those realms in which individuals cannot know all of the people with whom they interact or the enormous volume of factual knowledge that sustains their cooperation networks. This is to say that the ethics of care serves as a moral guide for the small group, whereas the ethics of justice serves as a moral guide for society. This reasoning can offer a sound foundation for some care ethicists' positions, like that of Held (2006), who thinks that the ethics of care should have priority in areas such as the family whereas the ethics of justice should have it in fields such as the law.

From this demarcation of the domains of the ethics of justice and the ethics of care, we can describe the interaction between the two ethics. But first we must say that our distinction based on factual knowledge must not be confused with the traditional private/public dichotomy. Unlike those who state that families are the only

[4] For the sake of exposition, we assume here that the small group is not embedded in the Great Society. It is true, as we mention below, that most small groups are currently embedded in the Great Society, but it is also true that some groups continue to live on the fringes of it, in the same way that many groups throughout history have not coexisted with the Great Society.

small groups, we understand the small group as any relationship with a known third party in the public or private sphere. Our contention is rather that the ethics of care can have priority both in relationships at work or neighborhood associations and in the family. Likewise, our argument does not hold that the ethics of care and the ethics of justice are mutually exclusive; rather, they may reinforce each other. We, like Ruddick (1995), see justice as working in tandem with care. This last idea leads us to address how the two ethics can interact in two different ways.

On the one hand, we posit that the ethics of care can help to ensure that people observe the rules of conduct contained in the ethics of justice that emerge spontaneously and thus ensuring that people contribute to the maintenance of social cooperation and the progress of civilization. Given that any current small group is embedded in the Great Society, its members must at the very least observe the abstract rules of conduct contained in the ethics of justice, although it may be that the primary moral guide they follow in the small group is that proposed by an ethics of care.

On the other hand, the fact that people understand that there is a comprehensive ethics of justice is not sufficient for ensuring that they observe rules of just conduct. It is indispensable that people are willing to respect the rules of the ethics of justice. That is why one traditional ethical approach, Kantianism, resorts to the idea of duty, which fills the gap between abstractly defined norms and real-life practice: you must obey rules because it is your duty.

From an ethics-of-care perspective, Held (2006, 71) thinks that care is "above and beyond the floor of duty." Caring for and about people is about more than complying with your obligations; it is about meeting their needs, knowing their backgrounds and histories, empathizing with them. In that sense, we can conceive another element that makes people observe rules of just conduct: people may think that these norms should be followed not only because of an abstract duty but because these rules enable millions of other people to make the most of their factual and personal knowledge (Hayek 1982)—that is, because they care for others. People embedded in an ethics of care look at abstract norms with empathy. The set of rules of just conduct allows people to live their lives in a complete way without harming the many they do not know. That is the only way to extend the values of the small group to the Great Society without altering practices conducive to cooperation. Therefore, by recognizing our irremediable ignorance, recognizing the need for rules of just conduct, and becoming invested in the preservation and observance of them, we care for the millions of unknown persons in society. In contrast with the Kantian idea of duty, this care-ethics alternative may represent a more powerful tool for fulfilling a morality of justice, provided that it makes more sense for people to comply with the rules because they know that they are beneficial for others than for them to recognize the rules as mere abstract duties.

Not only does the ethics of care reinforce the ethics of justice by making people comply with the rules of conduct out of empathy, but also the ethics of justice makes an ethics of care more effective. This is so because a Great Society functioning with abstract rules of conduct and a sophisticated division of labor is affluent, which gives people more means and leisure to care for others in small groups. Violation of the abstract rules of conduct destroys the material and psychological foundation for the ethics of care.

In this regard, we can find the most evident factor uniting the two ethics in cooperation. We cited Held (2006, 15) above where she states: "Whereas justice protects equality and freedom, care fosters social bonds and cooperation." This reasoning is based on the criticism by care ethicists of the individualistic values usually represented in the ethics of justice, such as autonomy or self-interest. However, the idea of the rational, autonomous agent is not supported by all of the authors who address how rules of conduct arise and ultimately how a morality founded on principles of justice emerges. For his part, Hayek (1982) rejects the rationalistic standpoint[5] and argues that rules of conduct and a sense of justice emerge spontaneously through cooperation and interactions. If cooperation is relevant for both ethics, how do they interact with each other? In order to answer this question, we must first realize that, even though rules of conduct arise spontaneously, they do not come from nothing but result from human actions.[6] The rules of the Great Society appear in the small group and are approved or discarded in an evolutionary process of trial and error in which the different small groups cooperate and interact. This has been the case historically, and it is so today.

As Hayek (1982) emphasizes, for a morality to work, something more than consistency or compatibility is required; namely, the morality must be able to produce a functioning order that maintains civilization. The system of rules that enables civilization to grow, corresponds to the sphere of the ethics of justice. Identifying which small-group ethics are consistent with the universal and abstract morality of justice allows us to learn whether such ethics are compatible with the expansion of civilization.

If we look in detail, the moral principles inspired by the practice of care that are suggested by care ethicists, such as responsiveness, responsibility, and meeting the needs of others (Held 2006), are principles that respect the impartial, abstract, and universal rules of conduct provided by the ethics of justice; by meeting the needs of others or caring for them, we are implicitly acknowledging their individuality and uniqueness, something that according to various care ethicists (Held 2006) is a central element in the ethics of justice. Such compatibility may lead us to think that a morality of care can be a source of a morality of justice.

A vast array of different moralities can be established in a small group. Many of them accept violence as a means; others see the whole as the only existing entity, in which the individual can be subordinated to the desire of the group or its leader. Yet, these kinds of norms are not consistent with the current morality of justice and thus are inconsistent with the growth of civilization. No small group could become a Great Society under a system of rules that does not allow people to use and expand

[5] The ethics of care is rooted in the feminist epistemological critiques of Cartesian rationalism. Feminist philosophers such as Bordo (1986) have argued that Cartesianism entailed the masculinization of science. According to Bordo, the rejection of subjectivity, emotions, and dependence—central concepts in the ethics of care—is a result of Cartesian rationalism. From another perspective, Hayek (1948, 1982) criticizes Cartesian rationalism for its erroneous assumption of the power of the human mind and introspection; he emphasizes limited human cognitive capacity, in contrast.

[6] More specifically, they are results of human action but not of human design, as Hayek (1982) remarks by citing Adam Ferguson's famous words.

the amount of factual knowledge. However, a morality of care seems compatible with the morality of justice, which brought society to the current state of prosperity. Caring entails recognizing other people's individuality—that is, recognizing that other people can freely use their factual knowledge. At the same time, caring goes beyond that recognition and mainly lies in the preservation of those people´s interests.

This argument may enable us to further argue, first, that it is possible that small groups whose rules were historically based on care became a Great Society before small groups whose morality was based on distinct values. This insight is supported by the high degree of compatibility between the morality of justice and the morality of care. And second, we argue that this process continues today, which means that in small-group situations in which the norms from the ethics of justice do not apply fully, the ethics of care can serve to establish consistent rules that are complementary with the current morality of justice and provide the basis for future universal rules that can be applied to the same issues at the Great Society scale. In the end, both lines of reasoning support the idea that the ethics of care can be the basis for the development of the ethics of justice (Held 2006).

Having expounded a new interpretation of the ethics of justice and ethics of care from the Hayekian standpoint of human knowledge limitations, we conclude that the ethics of care is compatible with a functioning order and the growth of civilization. Furthermore, we consider its role essential (1) for fulfilling a clearly defined morality of justice and (2) as a source in small groups of the further development of universal and abstract norms that fit into the sphere of the ethics of justice. To continue with our work, we present those two last points as a feasible ethical solution to the problem of the tragedy of the commons.

3 The ethics of care and the tragedy of the commons

The term "tragedy of the commons" was coined by the ecologist Garrett Hardin (1968). The term refers to situations in which individuals use a common resource, and assuming that these individuals act according to the rational-agent model—that is, assuming they act as utility-maximizing individuals. Utility-maximizing behavior leads to a tragedy: the overuse of the common resource. The most famous example that Hardin (1968) uses is a pasture open to all, such that each herder benefits from having her animals graze there but at the same time bears little cost from the resulting overgrazing. This ends in the exhaustion of the common resource, the pasture, as each herder tries to add more and more animals in order to maximize her gains.

Since Hardin (1968), tragedy-of-the-commons analysis has been applied to a wide range of situations. A shared characteristic of all of these applications is the absence of norms for using the common resource. To be more precise, this type of situation can occur for two reasons that we have previously expounded. First, in the absence of norms defined in principle, there is no norm at all. Second, despite the existence of defined norms, they might be not adhered to in practice, which, in the end, amounts to the same thing as not having norms. In fact, Hardin's (1968) work deals with the problem of overpopulation as a tragedy of the commons whose

solution needs to be found in morality, in a set of rules of conduct, rather than in technology.

If morality is a solution to the problem of the tragedy of the commons, how can the ethics of care contribute to this solution? In Sect. 2, we outlined how a morality of care can solve the problem of the tragedy of the commons. Although we did not mention the tragedy of the commons by name, we discussed the two possible situations that trigger the tragedy: (1) the non-observance of defined rules, and (2) the absence of defined rules. We suggested that a morality of care can alleviate or even eliminate the problem of non-observance of rules by serving as reinforcement for the observance of abstract and universal norms. And, we suggested that a proper morality of care can set rules in small groups that are compatible with a morality of justice and that lay the foundation for the development and improvement of universal rules. In order to make the point more evident and to illustrate our theory, we present two cases of tragedy of the commons: fractional-reserve banking and environmental tragedies.

3.1 Fractional-Reserve Banking

Fractional-reserve banking is a system in which banks do not hold in reserve 100 percent of customers' deposits but lend them and abide by a fractional reserve–deposit ratio. In contrast, in a 100 percent–reserve system, banks maintain full reserves (Abel et al. 2014). On the relative merits of these two alternative systems, there has been an intense debate in economic science. A central question is whether the fractional-reserve system causes business cycles.[7] Yet, the debate about fractional-reserve banking does not concentrate only on the system's economic consequences but also, and relevantly, on its ethical implications. Various economists have wondered whether the fractional-reserve system is ethically acceptable or, on the contrary, inadmissible in light of an ethics of justice (Bagus et al. 2018; Block & Davidson 2011).

Among the supporters of 100 percent–reserve banking, Huerta de Soto (2006) regards credit expansion in a fractional-reserve banking system and the subsequent economic collapse as a tragedy of the commons. The 100 percent–reserve banking theory (Rothbard 2000, 2001; Huerta de Soto 2006) shows that the practice of fractional-reserve banking—that is, the granting of loans and the creation of demand deposits unbacked by real savings—causes an artificial reduction of interest rates that motivates entrepreneurs to invest in more and longer projects than those that could be financed with the supply of real savings. Ultimately, a bust occurs as a result of the disruptive effects of banks' creation of loans on the structure of production.

[7] For all those interested in the debate, see Selgin (1988), White (1989), and Selgin and White (1996) for the opinion that fractional-reserve banking does not cause economic cycles. See also Hoppe (1994), Hoppe et al. (1998), Huerta de Soto (1995, 1998), and Bagus et al. (2013), who resume the criticism of fractional-reserve banking initiated by Mises (2009) and Hayek (2008).

Important from an ethical perspective is the fact that fractional-reserve banking alters depositors' preferences regarding their money. While the depositors want to have their money on deposit be completely available—otherwise, they would not use a deposit contract but would use a time-deposit contract—the depositary takes the money and lends it. That is the reason why depositors get upset when they do not have access to their money during a bank run. They find it scandalous that it is not available to them when they want it.[8] The depositors' real preference is not to invest this money but to have it held in custody with full availability. But, so long as the depositary is tempted to profit from lending the depositors' money, it may take that money and grant loans to third parties with it. In that process, the depositary artificially inflates the money supply and ceteris paribus reduces interest rates, thus feeding the credit expansion that finally ends in a recession in which the malinvestments are liquidated.

According to Huerta de Soto (2006), the problem of fractional-reserve banking derives from a legal and ethical corruption of the general principles of law, with a confusion between two different types of contracts: the irregular-deposit contract and the loan contract. In the former, the depositor transfers the property but not the availability of the *tantundem*,[9] whereas in the latter, the depositor transfers the ownership and availability of the *tantundem.* In the irregular-deposit contract, given the depositor's preference for having her goods lie in custody, the depositary is obligated to keep the goods available all of the time. A fractional-reserve contract for demand deposits entails a violation of the essential obligation to guarantee and keep always available the goods, which makes it an illegitimate and unethical practice. The violation of the safe-keeping obligation gives rise to an inadequate and vague definition of property rights to the goods deposited. Through fractional-reserve banking, banks issue titles to more property than is entrusted to them. They create the illusion that the money of all customers, both depositors and borrowers, is fully available, when in reality, it is not. As property rights are not adequately defined, it is not very clear who is the rightful owner.[10]

Following the above description of the business cycle's causes and the issues that fractional-reserve banking entails, Huerta de Soto (2006) applies the concept of the tragedy of the commons to fractional-reserve banking. First, he remarks that fractional-reserve banking involves an inadequate definition of property rights, which, as he affirms, is the crucial element of and trigger for the tragedy of the commons. As property rights on demand deposits are not clearly defined, the banker suffers the temptation to take the depositor's money and lend it to third parties. This granting of

[8] See also a survey conducted by Evans (2010) in which almost 90 percent of depositors state that the reason they keep deposits is to have convenient access to their money (and have it kept safe). For a discussion of the meaning of complete availability, see Bagus and Howden (2016).

[9] A Latin word that means the same quantity of goods of the same quality and type (Huerta de Soto 2006).

[10] From the perspective of current positive law, the bank is the owner of the deposits, but there is much confusion. In a survey (Evans 2010), only 8 percent of depositors were aware of the legislation; 74 percent thought they were the legal owners of "their" deposits.

loans implies the creation of new money and benefits the banker. After the bust, the costs spread throughout the economy in the form of a crisis.

The setup resembles a tragedy of the commons. Property rights on demand deposits are not clearly defined, and the practice of fractional-reserve banking allows banks to profit from expansionary policies while they bear only a little part (if any) of the external cost of the bust resulting from credit expansions. Banks can widely expand credit until the system cannot withstand any more crises. The ultimate limit of the exploitation of the commons is hyperinflation (Bagus 2004).

The fractional-reserves issue has been dealt with at the theoretical level, both in economic (Hoppe 1994; Hoppe et al. 1998; Huerta de Soto 1995, 1998; van den Hauwe 2008, 2010) and in legal and ethical terms (Bagus et al. 2013; Block & Davidson 2011; Huerta de Soto 2006; Hülsmann 2008; van den Hauwe 2006). Theory suggests that fractional-reserve banking is illegitimate and unethical and has adverse economic consequences. However, it stands out as the conventional system worldwide. How is that possible? In order to offer an answer to that conundrum, we combine our analysis of the tragedy of the commons in banking with the ethics of care. The tragedy of the commons occurs because of the non-observance of completely defined norms. Those norms, despite having emerged and having been observed during various historical periods, may have been abandoned because of lack of care, among other possible factors. This idea gains strength as we consider several arguments. For example, with the establishment of fractional-reserve banking, banks have deviated from meeting the needs and satisfying the preferences of their customers. Most depositors entrust their deposits to banks and expect complete availability (Evans 2010). This is their choice and desire; otherwise, they could sign a loan contract without complete availability that yields a certain return in a period. When a depositor wants to maintain the availability of his or her money, banks are thereby committed to *taking care of* the money. With fractional-reserve banking, they neglect not only their contractual obligation and a traditional legal principle but also the customer's preference and needs. Moreover, they do not care about the vast number of people damaged by the bust—the tragedy—they cause by expanding credit, if they are aware of the consequences at all. So, if, as we mentioned above, care entails sensitivity, meeting needs, or empathy, it is easy to see that these factors are neglected in the practice and underlying values of fractional-reserve banking. Banks neither take care of the money deposited nor care about their customers' needs and preferences or the people affected by the financial collapse. Their attitude seems to be just the opposite of what care would demand.

The lack of care for people is a consequence of lack of interest in observing abstract rules in this case. As people abandoned the traditional legal principles, credit expansion and bankruptcies resulted in harm to people. The solution to this problem is the establishment of a 100 percent–reserve banking system,[11] which would prevent the detrimental credit expansion and thus the tragedy of the commons

[11] Still, as Bagus (2010) proves, it is not sufficient to have a 100 percent–reserve requirement; it is also necessary that governments and central banks not bail banks out when they have roll-over problems or fail.

in banking (Rothbard 2001). It consists in re-establishing the observance of traditional legal principles. In order to achieve this solution, the values and practices delineated in the ethics of care may serve to bridge the gap between theory and practice—namely, to implement the abstract norms.

We posited that some traditional ethical approaches, such as Kantian ethics, attempt to bridge the gap between theory and real-life practice with the idea of duty. As care goes beyond duty, we showed how the ethics of care could successfully bridge that gap by offering potent reasons to comply with abstract norms, which is different from the traditional alternative of duty. The ethics of care looks to norms of empathy that extend beyond obligation and support universal rules that allow for the development of millions of persons. This is precisely the element that could make it possible for people to comply with the rules of a morality of justice, thus bridging the gap between theory and practice more effectively than, for example, Kantian ethics does.

The subject of fractional-reserve banking illustrates this thesis. It might help if banks complied with the abstract rule of keeping 100 percent reserves for demand deposits, aware that their activity would thereby not end in the tragedy of the commons the business cycle represents. It might help if banks realized that, in observing traditional legal principles, they would be meeting the needs of their customers, taking care of the money deposited, caring about their customers' preferences and desires, and caring about those harmed by the bust. Therefore, if banks embrace a morality of care, they may resume compliance with traditional legal principles, avoiding a tragedy of the commons.[12] By the same token, as we previously demonstrated, it is precisely the lack of a morality of care that has caused the recurrent tragedy of the commons related to the banking system.

3.2 Environmental Tragedy

In contrast to the tragedy of the commons resulting from fractional-reserve banking, the tragedy involved in the use of environmental resources has already been explained; in fact, it was explained in Hardin (1968) itself. Since property rights to environmental resources are not clearly defined, considerable benefit goes to those who exploit the resources while the cost is born by many people, which produces incentives to overuse the resource and exhaust it. In this case, the tragedy is triggered by the presence of neither abstract nor concrete rules for managing such a situation in a small group.

A great deal of literature on environmental resources management has treated the problem noted by Hardin (1968), one of the most well-known contributions being that made by Nobel Prize winner Elinor Ostrom (1990). Her proposal to solve the problem of the tragedy of the commons is of great interest in light of the ethics of care. She proposes a solution different from the two traditional alternatives. The traditional alternatives according to Ostrom consist, on the one hand, in the

[12] We have to admit that banks not holding full reserves have a competitive advantage. In any case, the prevalence of an ethics of care would increase the psychic costs involved in credit expansion.

management of the resource by a central agency using its coercive tools and, on the other hand, in the total privatization of the resource. She argues that both options, the market and the state, fail to reach an optimal point. Moreover, she criticizes the idea that the solution must come from an imposition by an external actor. She does not believe in a rigid and exact answer but in many different solutions for many different problems. Ostrom (1990) focuses on how individuals in diverse communities—small groups—manage their common-pool resources[13] (CPRs) in an attempt to avoid tragedies, whether successfully or not.

In our discussion on fractional-reserve banking, we treated a situation in which individuals and organizations neglect to observe clearly defined rules in practice, causing a tragedy of the commons. We have yet to examine the second possibility, which is about the non-existence of rules in a small group and how a morality of care can be the source of rules that prevent the tragedy and how it can even lay the foundations for the transformation of the rules of the small group into the Great Society's rules. Ostrom delves into a scenario which is precisely the one we have yet to address: small groups lacking rules. Ostrom (1990) rejects two Great Society–scale options for the management of CPRs, both the state option and the market one, as she opposes the idea that the solution has to come from impositions by an external actor. Instead, she prefers to study how the individuals in a group self-organize, developing their own rules to reach a solution. Ostrom (1990, 51) makes it clear:

In many CPR settings, the working rules used by appropriators may differ considerably from legislative, administrative, or court regulations. The difference between working rules and formal laws may involve no more than filling in the lacunae left in a general system of law. More radically, operational rules may assign de facto rights and duties that are contrary to the de jure rights and duties of a formal legal system. My primary focus in this study will be on the de facto rules actually used in CPR field settings, in an effort to understand the incentives and consequences they produce.

After examining several successful and failed cases of management of CPRs in different groups around the world, Ostrom (1990) identifies a set of similarities, which she calls design principles, that make a CPR institution effective, robust, and enduring. These design principles are not concrete rules but necessary features that, following Ostrom (1990, 90), "help to account for the success of these institutions in sustaining the CPRs and gaining the compliance of generation after generation of appropriators to the rules in use." However, the essential point, in our view, resides in the moral values that underlie each institution. Without these values no kind of organization would be possible, since there would be no cooperation. In fact, the unsuccessful CPR cases failed not only on account of bad institutional arrangements, but also because of the absence of those moral values that permeate each institution and constitute an indispensable requirement for its success and endurance. Ostrom

[13] Ostrom (1990, 30) defines a CPR as "a natural or man-made resource system that is sufficiently large as to make it costly (but not impossible) to exclude potential beneficiaries from obtaining benefits from its use."

(1990) refers to values such as trust, sense of community, integrity, reputation as reliable group members, communication, and reciprocity. From an ethics-of-care perspective, it is not difficult to identify all of these values in the practice of care. We focus on trust because it is a crucial value for Ostrom (1990) and also for some care ethicists. Baier (1994) and Held (2006) think that trust is central to morality. Although caring is not the same thing as trust, they sustain each other. In the end, trust is the basis of social cooperation that allows small groups to define institutional arrangements to avoid the tragedy of the commons.

Trust can be embedded in a large variety of small groups' moralities, and it fosters a morality of care. Conversely, a morality of care can best integrate trust and promote its value and therefore facilitate better than other moralities the kind of social cooperation that avoids the tragedy of the commons. That is so because from the ethics-of-care perspective, trust is more than an individual virtue or the basis for predicting a third party´s behavior. It is a mutual understanding about people's intentions. Plus, caring and trust go hand in hand to create a caring community in which people need to trust others in order to respond to their needs. Caring is how trust is made concrete in many different respects (Held 2006). That conception goes beyond the idea of merely creating a climate of trust which helps people face uncertainty, and so it may attain social cooperation and a solution to the tragedy of the commons more effectively. Furthermore, for care ethicists, care extends beyond the mere fact of caring for others; it also encompasses care for the environment (Petit 2014), which reinforces our argument that the morality of care is a cooperative solution to the tragedy of the commons. A morality of care explicitly entails care for the environment, besides implicitly entailing the care derived from a successful institutional arrangement built on the values that the ethics of care advocates.

To summarize our argument, the underlying values required for successful CPR institutions, especially trust, are compatible with and similar to the values inherent in a morality of care. Therefore, we maintain that the solution to the tragedy of the commons, following Hardin (1968), rests on an effective morality, such as a morality of care, to guide the actions of people in a community so that they can build successful institutions for the management of CPRs.

At this point, we have to deal with the question of how small-group rules influence and, at the same time, evolve into Great Society rules. To this end, we refer to the work of Hayek. There are plenty of similarities between Ostrom´s and Hayek´s works. For example, Ostrom's (1990) argument against central planning as a way to manage CPRs is the same that Hayek (1945) made against any kind of central plan. By the same token, she refers to rules of diverse scales (operational, collective-choice, and constitutional-choice rules) and asserts that all rules are embedded in other sets of rules, where higher levels of rules depend on hard-to-change deeper levels of rules. Ostrom's (1990) thought is very close to the theory about small-group and Great Society rules that we expounded at the beginning of this article. The Hayekian approach also supports the existence of different levels of rules depending on the factual knowledge they encompass. Thus we have small-group rules, which cover an amount of factual knowledge that allows them to be rationally designed, and Great Society rules, which concern such a vast amount of factual knowledge that it is not possible to design them rationally. As is obvious, the first

rules that appeared were small-group rules, which, in the process of civilization, led to the emergence of Great Society rules. We affirmed that this process keeps going today. This raises the question of how small-group rules can influence and evolve into Great Society rules.

Let us begin with a critique of Ostrom. Block and Jankovic (2016) emphasize two errors in Ostrom (1990). First, they assert that she misunderstands the concept of property rights when she alludes to the commons and its governance as an alternative to the central-planning and property rights approaches, whereas she is actually talking about diverse forms of private property, which include partnerships and condominiums. Partnerships and condominiums are not at odds with the market but part of it. Second, Block and Jankociv (2016), providing some historical evidence, demonstrate that government is not necessary to regulate or enforce property rights, and they thereby give support to the idea that governance of the commons is a form of private property. They inspect Ostrom's (1990) theory in search of logical inconsistencies with traditional legal principles such as those of Roman law or common law. Traditional legal principles are on the Great Society scale. In this regard, Block (1990, 1998) argues against central planning and in defense of the free market solution for environmental problems; that is, he recommends establishing private property rights to each environmental resource. He does not offer concrete operational rules or strategies but demonstrates that government intervention can never internalize externalities and avoid the tragedy of the commons. The reasoning is that, given that government control always fails to avoid the tragedy, market participants should coordinate with each other and manage all of the environmental resources. How each group or community manages their resources only concerns them, the owners; it is a matter of small-group rules.

With Block (1990, 1998), Block and Jankovic (2016), and Ostrom (1990), we see how the issue of the tragedy of the commons in the environment, as with any other tragedy of the commons, can be treated at both the small-group and Great Society scales. The two standpoints can complement each other. While the rules of the Great Society constitute a basic and universal standard that serves to regulate cooperation and relationships among all communities and individuals that coexist in the Great Society, the small-group rules specify the operational rules that determine the success or failure of a particular institutional arrangement. There is an interaction between the ethics of care and the ethics of justice. The Great Society did not appear out of nowhere. Traditional legal principles, private property rights, and condominiums resulted from a long evolutionary process, which involves diverse small groups with their own rules and moralities cooperating and competing. Mousourakis (2012, 1) points out that.

Roman law was devised for a small, rural community that developed into a powerful city-state and it evolved as the law of a multinational empire that embraced a large part of the civilized world. During this long process the interaction between custom, enacted law and case law led to the formation of a highly sophisticated system gradually developed from layers of different elements.

In the case of condominiums, a successful Roman CPR management practice gave rise to the institutional arrangement called a condominium—that is, a form of joint ownership. In particular, as Mousourakis (2012) explains, the *communion*

pro indiviso, or condominium, originated when two or more individuals acquired through legacy the same property in common and voluntarily decided to share it, subject to certain norms. What if those individuals had not been guided by trust, reciprocity, or even care—given that most cases of common property arose from inheritances in the sphere of the family, in which care is supposed to reign, in contrast to other realms—and had resorted to violence and conflict as antagonists? Neither a successful CPR institution nor its Great Society counterpart, the condominium, would have appeared. With these historical references, we illustrate our theory about the interaction between Great Society rules and small-group rules, which is the same as saying the interaction between the ethics of justice and the ethics of care. Great Society rules emerge from small-group rules, and once both coexist, they reinforce each other, and the small-group rules modify the Great Society rules in an evolutionary process (Hayek 1982).

From a static point of view, it should be clear that the ethics of care, which we previously confined to the realm of the small group, cannot deal with the moral problems of the Great Society, and vice versa. Each established ethic has its sphere of application. However, our theory, supported by historical evidence, demonstrates that from a dynamic point of view, Great Society rules depend on small-group rules. This being so, a morality of care becomes very relevant since it promotes values that facilitate cooperation and the development of complex social processes, which, in the spheres of both the small group and the Great Society, lay the foundations for the solutions to different tragedies of the commons.

4 Conclusion

In this paper, we have shown how the ethics of care can offer solutions to diverse tragedies of the commons. To do so, we first resorted to Hayek's (1982) theory on the emergence of rules and norms in order to demarcate the spheres of application of the ethics of care and the ethics of justice and to outline how the morality of care interacts with the morality of justice. The ethics of care is confined to moralities of small groups, whereas the ethics of justice must apply to moralities of the Great Society, given the limitations of human knowledge. Likewise, a morality of care can bridge the gap between theory and practice at the Great Society scale, achieving the observance of a morality of justice when it was not being observed, and it motivates the creation of successful small-group rules that evolve into Great Society rules over time.

We affirmed that a morality of care allows people to establish rules in small groups and to observe theoretically defined rules in the Great Society, thus avoiding any tragedy of the commons in either realm. We illustrated the point with two typical examples of the tragedy of the commons: fractional-reserve banking and environmental tragedies. From a dynamic point of view, the solution to each tragedy is to be found in a morality of care.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

References

Abel AB, Bernanke BS, Croushore D (2014) Macroeconomics, 8th edn. Pearson, New York

Bagus P (2004) La tragedia de los bienes comunales y la escuela austriaca: Hardin, Hoppe, Huerta de Soto y Mises. Procesos De Mercado: Revista Europea De Economía Política 1(2):125–134

Bagus P (2010) Austrian business cycle theory: are 100 percent reserves sufficient to prevent a business cycle? Libertarian Papers 2(2):1–18

Bagus P, Howden D (2016) The economic and legal significance of "full" deposit availability. Eur J Law Econ 41:243–254

Bagus P, Howden D, Block W (2013) Deposits, loans, and banking: clarifying the debate. Am J Econ Sociol 72(3):627–644

Bagus P, Gabriel A, Howden D (2018) On the necessary and sufficient conditions for legitimate banking contracts. J Bus Ethics 147(3):669–678

Baier AC (1994) Moral prejudices: essays on ethics. Harvard University Press, Cambridge

Barry B (1995) Justice as impartiality. Oxford University Press, Oxford

Block W (1990) Economics and the environment: a reconciliation. The Fraser Institute, Canada, Vancouver

Block W (1998) Environmentalism and economic freedom: the case for private property Rights. J Bus Ethics 17:1887–1899

Block W, Davidson L (2011) The case against fiduciary media: ethics is the key. J Bus Ethics 98:505–511

Block W, Jankovic I (2016) Tragedy of the partnership: a critique of Elinor Ostrom. Am J Econ Sociol 75(2):289–318

Bordo S (1986) The Cartesian masculinization of thought. Signs: J Women Culture Soc 11(3):439–456

Bubeck DE (1995) Care, gender, and justice. Oxford University Press, Oxford

Colombo M (2014) Caring, the emotions, and social norm compliance. J Neurosci Psychol Econ 7(1):33–47

Evans, A. J. (2010) Public attitudes to banking. A student consultancy project. *ESCP Europe for the Cobden Centre.*

England P (2005) Emerging theories of care work. Ann Rev Sociol 31(1):381–399

Gilligan C (1982) In a different voice: psychological theory and women's development. Harvard University Press, Cambridge

Hardin G (1968) The tragedy of the commons. Science 162(3859):1243–1248

Hauwe van den L (1998) Evolution and th production of rules - some preliminary remarks. Eur J Law Econ 5:81–117

Hauwe van den L (2006) The uneasy case for fractional-reserve free banking. Procesos De Mercado Revista Europea De Economía Política 3(2):143–196

Hauwe van den L (2008) Credit expansion, the prisoner´s dilemma, and free banking as mechanism design. Procesos De Mercado Revista Europea De Economía Política 5(2):133–174

Hauwe van den L (2010) Free banking, the real-balance effect, and Walras´law. Procesos De Mercado Revista Europea De Economía Política 7(1):241–258

Hayek FA (1945) The use of knowledge in society. Am Econ Rev 35(4):519–530

Hayek FA (1948) Individualism: true and false. In: Hayek FA (ed) Individualism and economic order. The University of Chicago Press, Chicago, pp 1–32

Hayek FA (1982) Law, legislation and liberty. Routledge, Abingdon

Hayek, F. A. (2008) *Prices and production and other works: F.A. Hayek on money, the business cycle, and the gold standard.* Auburn: The Ludwig von Mises Institute.

Held V (2006) The ethics of care: personal, political, and global. Oxford University Press, New York

Hoppe H-H (1994) How is fiat money possible?-or, the devolution of money and credit. The Review of Austrian Economics 7(2):49–74

Hoppe H-H, Hülsmann JG, Block W (1998) Against fiduciary media. Q J Austrian Econ 1(1):19–50

Huerta de Soto J (1995) A critical analysis of central banks and fractional-reserve free banking from the Austrian school perspective. The Review of Austrian Economics 8(2):25–38
Huerta de Soto J (1998) A critical note on fractional-reserve free banking. Q J Austrian Econ 1(4):25–49
Huerta de Soto J (2006) Money, bank credit, and economic cycles. The Ludwig von Mises Institute, Auburn
Hülsmann JG (2008) The ethics of money production. The Ludwig von Mises Institute, Auburn
Kittay EF (1999) Love's labor: Essays on women, equality, and dependency. Routledge, New York
Leoni B (1991) Freedom and the law, 3rd edn. Liberty Fund, Indianapolis, In
Liedtka JM (1996) Feminist morality and competitive reality: A role for an ethic of care? Bus Ethics Q 6(2):179–200
Mises L (2009) The theory of money and credit. The Ludwig von Mises Institute, Auburn
Mousourakis G (2012) Fundamentals of Roman private law. Springer, New York
Nelson JA (2011) Care ethics and markets: A view from feminist economics. In: Hamington M, Sander-Staudt M (eds) Applying care ethics to business. Springer, New York, pp 35–53
Nelson JA, England P (2002) Feminist philosophies of love and work. Hypathia 17(2):1–18
Noddings N (1984) Caring, a feminine approach to ethics and moral education. University of California Press, London
Ostrom E (1990) Governing the Commons. the evolution of institutions for collective action. Cambridge University Press, Cambridge
Paillé P, Mejía-Moleros JH, Marché-Paillé A, Chen CC, Yang C (2016) Corporate greening, exchange process among co-workers, and ethics of care: an empirical study of the determinants of pro-environmental behaviors at coworkers-level. J Bus Ethics 136:655–673
Petit E (2014) Ethique du care et comportement pro-environnemental. Revue D'economie Politique 124(2):243–267
Pulcini E (2013) Care of the world: fear, responsibility and justice in the global age. Springer, New York
Rothbard MN (1982) The ethics of liberty. Humanities Press, Atlantic Highlands
Rothbard MN (2000) America's great depression. The Ludwig von Mises Institute, Auburn
Rothbard MN (2001) The case for a 100 percent gold dollar. The Ludwig von Mises, Auburn
Ruddick S (1995) Injustice in families: Assault and domination. In: Held V (ed) Justice and care: essential readings in feminist ethics. Westview Press, Boulder, pp 203–223
Selgin GA (1988) The theory of free banking: money supply under competitive note issue. Rowman and Littlefield, Totowa
Selgin GA, White LH (1996) In defense of fiduciary media-or, we are not devo(lutionists), we are misesians! Review of Austrian Economics 9(2):83–107
Solomon RC (1998) The moral psychology of business: Care and compassion in the corporation. Bus Ethics Q 8(3):515–533
Tronto JC (1993) Moral boundaries: a political argument for an ethic of care. Routledge, New York
van Staveren I (2005) Modelling care. Rev Soc Econ 63(4):567–586
van Staveren I (2013) Caring finance practices. J Econ Issues 47:419–425
White LH (1989) Competition and currency. New York University Press, New York
White TI (1992) Business, ethics, and Carol Gilligan's "two voices." Bus Ethics Q 2(1):51–61

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